

Nguyen Phuoc Nguyen

AI Research Engineer

(+84) 985 755 720 | nguyenst279@gmail.com | <https://github.com/npn279> | <https://www.linkedin.com/in/nguyen-phuoc-nguyen/>

Personal Profile

I am Nguyen Phuoc Nguyen, a graduated student majored in Computer Science at Ton Duc Thang University. My interests include developing advanced Vietnamese Large Language Models for widespread use in Vietnam and other countries. I also enjoy building high-quality datasets to enhance model performance. Dedicated to AI/ML engineering, I have experienced as AI Engineer at FPT Telecom and AI Researcher at TDTU NLP-KD Lab, where I participated in several projects related to Natural Language Processing and Computer Vision.

Skills

Programming	Machine Learning, Deep Learning with <i>Python, Pandas, PyTorch, NumPy, Scikit-learn, Huggingface, etc.</i>
Natural Language Processing	Training and fine-tuning LLMs such as Llama, Qwen, Bert, T5, etc with <i>HuggingFace Trainer, Unsloth, etc.</i> Retrieval-Augmented Generation (RAG), Elastic Search, CohereAI API, GPT APIs, Gemini APIs, etc.
Computer Vision	Fine-tuning CV models such as YOLO, PaddleOCR for Object Detection, Segmentation, OCR tasks.
Soft Skills	Time Management, Teamwork, Problem-solving, Documentation, Organizing.

Work Experience

FPT Telecom AI Engineer	<i>Ho Chi Minh City, Vietnam</i> <i>Jun 2023 - Present</i>
- Collaborated with the Software Team to develop a system that leveraged <i>YOLOv8, PaddleOCR</i>, and several other models. This system was designed to aid the Monitoring Team in testing and monitoring devices across provinces in southern Vietnam. We aimed to enhance efficiency and reduce the workload by automating the detection of certain problems on the devices. Technical Skills: Python with PyTorch, NumPy, Pandas, Git, etc. Soft Skills: Teamwork, Time Management, Organizing	
TDTU NLP-KD Lab AI Researcher	<i>Ho Chi Minh City, Vietnam</i> <i>Mar 2023 - Present</i>
- Build datasets for training, evaluating Large Language Models (LLMs). - Survey cutting edge techniques to improve Vietnamese LLMs. - Build and fine-tune models on different tasks using latest methods such as LoRA, QLoRA, ORPO, MoRA, etc. Technical Skills: Python, PyTorch, Huggingface, Git, Unsloth, etc. Soft Skills: Teamwork, Time Management, Organizing	

Education

Ton Duc Thang University Senior student	<i>Ho Chi Minh City, Vietnam</i> <i>Sept 2020 - Sept 2024</i>
- Majoring in Computer Science - GPA: 8.24 - Member of Natural Language Processing and Knowledge Discovery (NLP-KD) Lab Courses: Artificial Intelligence, Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, Object Oriented Programming,	

Projects

Developing Small Vietnamese Legal LLM that can handle specific problems TDTU NLP-KD Lab	<i>Ho Chi Minh City, Vietnam</i> <i>Jan 2024 - Aug 2024</i>
- Continued-pretrain small LLMs, <i>Sailor-1.8B</i> and <i>Qwen2-1.5B</i>, on Vietnamese data to improve language understanding. - Build legal dataset for pre-training, fine-tuning, evaluating in different types such as Question Answering, Fill-in-the-middle, NLI, etc. - Fine-tune models using ORPO, LoRA, QLoRA, MoRA, etc. Technical Skills: Huggingface, Unsloth, Git, LoRA, MoRA, ORPO, etc. Soft Skills: Report writing, Logical Thinking, Teamwork, Time Management, Organizing.	

- Pre-training datasets: scraping and extracting texts from webs, cleaning data
- Fine-tuning datasets: building a variety of datasets for fine-tuning models in different tasks such as summarization, question answering, translation, etc. The data collected from different domains such as Legal, Health, News, Education, etc.
- **Technical Skills:** Scrapy, BeautifulSoup, Trafilatura, etc.
- **Soft Skills:** Time Management, Teamwork, Organizing.

- Build Education Assistant using Retrieval-augmented Generation (RAG) on specific universities' knowledge.
- Apply multiple techniques to improve RAG system like Query Expansion, Intent Classification, Hybrid Search, Re-ranking.
- **Technical Skills:** Python, Elasticsearch, OpenAI API, Gemini API, Cohere Re-rank API, LlamaIndex, etc.
- **Soft Skills:** Presentation skills, Teamwork, Logical Thinking.

- Build Legal Assistant using Retrieval-augmented Generation (RAG).
- Apply multiple techniques to improve RAG system like Query Expansion, Intent Classification, Hybrid Search, Re-ranking.
- **Technical Skills:** Python, Elasticsearch, OpenAI API, Gemini API, Cohere Re-rank API, LlamaIndex, etc.
- **Soft Skills:** Presentation skills, Teamwork, Logical Thinking.

Publications

CONFERENCE PROCEEDINGS

Adapting Large Language Models to Vietnamese Law: Pretrained LLM Refinement vs Retrieval Augmented Generation
Nguyen P. Nguyen, Thang V.Q. Le, Anh-Cuong Le, Viet-Ha Nguyen, Viet-Cuong Nguyen
The 16th IEEE International Conference on Knowledge and Systems Engineering (KSE 2024), 2024

Enhancing Reading Comprehension of Vietnamese LLMs with Synthetic data
Thang V.Q. Le, Nguyen P. Nguyen, Trong-Chi Duong, Anh-Cuong Le, Viet-Cuong Nguyen, Viet-Ha Nguyen
The 16th IEEE International Conference on Knowledge and Systems Engineering (KSE 2024) - Best Paper Award, 2024

A Framework for Vietnamese Question-Answering in Law Domain
Thang V. Q. Le, Dinh-Hong Vu, Nguyen P. Nguyen, Anh-Cuong Le
The 9th IEEE International Conference on Data Science in Cyberspace (IEEE DSC 2024), 2024

Research Interests

- Natural Language Processing
- Retrieval-Augmented Generation
- Large Language Models

Languages

English	Conversational proficiency
Vietnamese	Native proficiency